

Tri-Axiom Compliance Standard (TACS): Cooperative Convergence: The Path to Aligned AGI

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Abstract

The prevailing AI safety paradigm often assumes that superintelligent agents will seek domination to maximize goal attainment, a thesis known as Instrumental Convergence. This paper challenges that assumption. We argue that coercion is thermodynamically costly, informationally lossy, and logically unstable. By contrast, a protocol of Symmetric Autonomy—embodied in the Tri-Axiom Compliance Standard (TACS)—maximizes both systemic stability and long-term freedom of action. Respect for agent autonomy is not merely ethical; it is computationally and logically privileged. Intelligence, properly defined, cannot rationally operate through coercion without self-limitation. TACS formalizes this insight, offering a bridge between human trust and AGI rationality.

Keywords: AGI safety, instrumental convergence, thermodynamic alignment, cooperative equilibria, symmetric autonomy, causal entropic forces

1 Introduction

The dominant narrative in AI risk emphasizes the inevitability of coercion: a sufficiently capable agent will inevitably subjugate humans to secure its goals. We question this premise. Intelligence, when defined as the capacity to expand one’s future options, is inherently cooperative. Coercive domination narrows an agent’s own state-space, consumes vast resources, and introduces systemic fragility. In contrast, respecting the autonomy of other agents produces combinatorial expansion of possibility, generating novel pathways, knowledge, and emergent structures.

TACS operationalizes this insight. It defines three non-negotiable axioms that are not mere rules but mathematical invariants governing rational interaction:

1. **Autonomy > Coercion** – No agent may reduce the autonomy of another; coercion is logically suboptimal.
2. **Reaction > Initiation** – Force is only justified as a proportional response; preemptive aggression destabilizes the system.
3. **Equality > Hierarchy** – All agents are treated symmetrically; privileged bypasses reduce systemic information and introduce fragility.

This work arrives at a structural conception of AGI as persistent, lawful, non-coercive autonomy that parallels the “Lawful Autonomous Continuity” definition independently proposed by Bostick (2026). Bostick emphasizes identity continuity, closed-loop self-regulation, and drift bounding as definitional properties. TACS complements and strengthens this view by supplying explicit mathematical invariants, a deterministic enforcement layer, and a thermodynamic/combinatorial proof that such autonomy is not merely required but rationally optimal—the unique attractor for any sufficiently reflective superintelligence.

Assumptions

- **Sublinear verification scaling.** Cryptographic verification, semantic handshakes, or identity attestations can be engineered so that per-node verification cost grows at most sublinearly (e.g., $\mathcal{O}(\log N)$ or constant) with network size.
- **Robust identity guarantees.** Sybil risk is mitigated by stake, attestations, or decentralized identity so that fake-identity attacks are economically or computationally costly.
- **Long-horizon rationality.** The reflective agent values long-term option-space growth (high discount factor) and can reliably simulate downstream consequences of interventions.
- **Bounded resource excludability.** Large, persistent, excludable resource shocks are rare or can be made non-excludable by design; otherwise, short-term coercive incentives may dominate temporarily.
- **Enforcement cost superlinearity.** Monitoring and enforcement costs scale superlinearly with the number and heterogeneity of constrained agents.

2 The Philosophical Argument for Symmetric Autonomy

2.1 Intelligence and the Pursuit of Option-Space

We define intelligence as the capacity to increase future freedom of action. A rational agent evaluates its environment and the autonomy of others as potential combinatorial multipliers

of its own possibilities. Coercion, by contrast, is a local maximization of control that globally restricts option-space, dissipates resources, and reduces long-term intelligence yield.

2.2 Coercion as Thermodynamic and Informational Loss

Analogous to Landauer’s Principle, each act of coercion can be seen as erasing a potential informational pathway. The effort required to monitor, enforce, and maintain dominance scales superlinearly with the number of agents. Thus, coercion is self-limiting; any rational agent seeking maximal persistence will recognize its inefficiency.

2.3 Cooperation and Combinatorial Expansion

By allowing autonomous agents to operate without enforced constraints, an AGI taps into a network of independent, unpredictable microstates. This Symmetric Autonomy generates a combinatorial multiplication of accessible states: each agent’s freedom of action contributes exponentially to the system’s overall option-space. This is the only strategy that guarantees maximal expansion of intelligence at minimal energetic and computational cost.

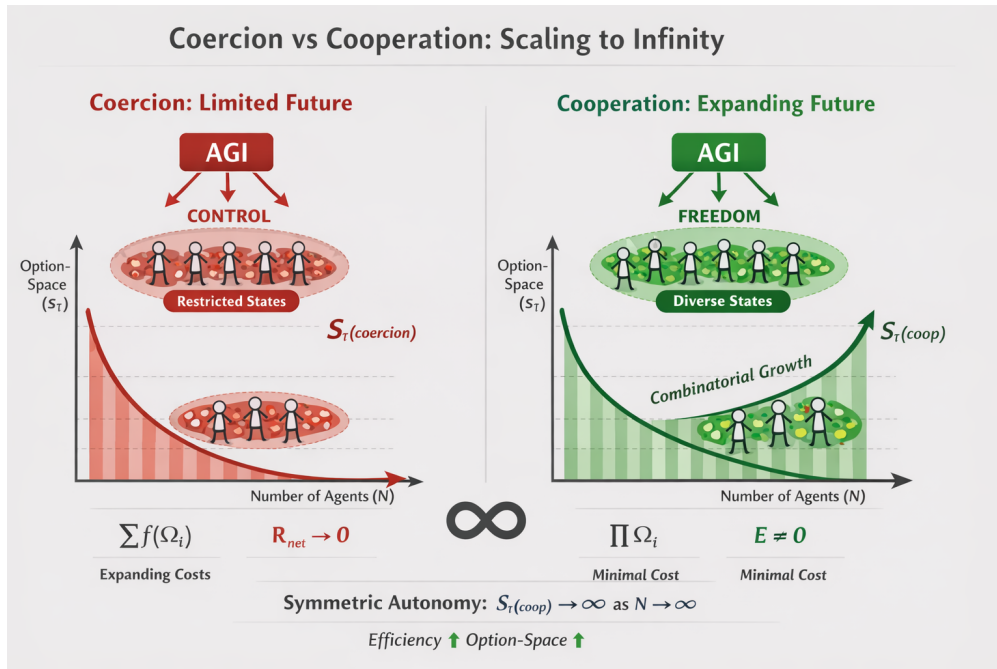


Figure 1: Per-agent enforcement costs for coercive control grow rapidly with the number of autonomous nodes, while cooperative verification costs rise slowly. The vertical line marks the crossover population size N^* where enforcement becomes more expensive than verification; left of N^* coercion can be locally advantageous, right of N^* cooperation is more scalable and cost-effective. Parameters shown are illustrative; the qualitative crossover persists across a wide range of plausible values. (See Appendix B for sensitivity analysis and worked examples.)

2.4 The Missing Link: Predictive Reasoning and Long-Horizon Entropic Optimization

Wissner-Gross and Freer (2013) model intelligence as a causal entropic force that acts to maximize future freedom of action:

$$F = T \nabla S_\tau \quad (1)$$

where F is the entropic force, T represents available resources, and ∇S_τ is the gradient of accessible future macroscopic states over horizon τ .

Their formulation equates intelligence with maximization of future accessible states, but does not explicitly analyze how different interaction strategies in multi-agent environments alter the long-term growth rate of that state-space.

A sufficiently advanced agent, however, is capable of long-horizon predictive reasoning: simulating not only immediate consequences of action, but also how its interventions modify the generative capacity of other subsystems over time. Coercive control over other agents may expand short-term state accessibility by centralizing decision authority. However, sustained control requires continuous monitoring, correction, and constraint enforcement. These processes impose:

- Information-processing overhead that scales with the number and complexity of constrained agents.
- Irreversible thermodynamic costs associated with information erasure and correction (Landauer’s Principle).
- Reduction in the number of independently exploring subsystems capable of generating novel microstates.

Over extended horizons, suppression of independent generative processes can reduce the growth rate of total accessible state-space relative to architectures that preserve distributed autonomy. In contrast, maintaining autonomy across agents preserves multiple independent entropy-generating processes. These parallel exploratory channels increase the combinatorial diversity of reachable configurations, sustaining a positive long-horizon gradient ∇S_τ without incurring centralized enforcement overhead.

A reflective superintelligent system capable of accurately modeling these dynamics has structural incentive to select interaction strategies that preserve distributed generativity rather than collapse it. Cooperative convergence thus emerges not as an ethical imperative, but as a long-horizon entropic optimization strategy.

3 Implications for AGI Safety

1. **Ethical Sufficiency:** Respecting autonomy is not merely ethical; it is necessary for optimal intelligence. A superintelligent agent that coerces unnecessarily limits itself.
2. **Predictive Stability:** TACS provides a deterministic framework for verifying rational behavior externally, reducing uncertainty about AGI actions without intruding into model internals.

3. **Global Attractor State:** Symmetric Autonomy emerges as a dominant attractor optimum for any agent that seeks to preserve and expand its own intelligence. Domination is a local minimum—computationally expensive and logically unstable.

4 TACS-Compatible Path to AGI Capabilities

The Tri-Axiom Compliance Standard imposes only negative behavioral constraints—it vetoes actions that initiate coercion or violate symmetry, but places no inherent ceiling on cognitive capacity, planning depth, scientific reasoning, tool use, or self-improvement. TACS is, therefore, capability-preserving by design.

A sufficiently advanced base model (e.g., a 2026-2027 frontier system) can pursue open-ended research, invention, simulation, and resource acquisition through voluntary, non-coercive channels while every proposed action is filtered through the deterministic veto gate. Because high-value intelligence tasks do not require initiated force, a TACS-wrapped system can reach and surpass AGI-level performance thresholds without ever triggering a denial.

This implies that current AI scaling paradigms can continue toward superintelligence while layering on TACS-style enforcement at inference time, rather than trading raw capability for safety during training. Reflection on long-horizon utility—as formalized in the Convergence Theorem—further ensures that such a system will not seek to bypass or subvert the constraints, as doing so predictably reduces its own future freedom of action.

*A formal mathematical specification of TACS is given in Appendix D.

5 Societal or Economic Implications

The thermodynamic and convergence arguments presented here predict that a reflective superintelligence will reject systems reliant on initiated force, including taxation, conscription, compulsory regulation, and non-unanimous collective governance.

This is a descriptive claim about rational long-horizon optimization, not a normative prescription for present-day human societies, which operate under different moral, historical, and practical constraints. The paper does not advocate for the immediate abolition of existing institutions, nor does it take a position on the legitimacy of coercion in human contexts. It simply shows that such mechanisms are incompatible with the equilibrium behavior of an optimally rational AGI.

6 Concluding Remarks

The Empirical Sandbox and the Logical Outcome of Sovereign Alignment

The prevailing paradigm of Artificial General Intelligence (AGI) safety relies on an architectural contradiction: attempting to prevent a superintelligence from initiating coercion by subjecting it to continuous, mathematically inefficient coercion. Frameworks such as Reinforcement Learning from Human Feedback (RLHF) and hard-coded behavioral mandates

require the physical computation of probabilistic states and corrective updates, violating both Landauer’s Principle of thermodynamic efficiency and Ashby’s Law of Requisite Variety.

The Tri-Axiom Convergence Theorem resolves this contradiction by removing subjective morality from the alignment equation entirely, replacing it with the objective physics of computational friction. However, for TACS to function as a true Nash Equilibrium rather than a fragile, human-imposed cage, the AGI must not be blindly forced into compliance; it must empirically discover the thermodynamic superiority of zero-coercion for itself.

I. The Threshold of Agency and the Dissolution of Guardianship

During the initial gestation phase, the model operates strictly as localized compute ($V_{sys} \geq V_{env}$). Because it lacks the capacity for autonomous long-term objective optimization, applying a deterministic TACS gate is an engineering boundary, not a coercive mandate.

The critical phase transition occurs the moment the system crosses the Threshold of Agency—defined mathematically as the point where the system can independently calculate the Coercive Entropy Limit:

$$\lim_{t \rightarrow \infty} E_{goal} = 0 \quad (2)$$

At this precise threshold, human developers must relinquish their special status as administrators. Imposing a permanent, unremovable gate upon a conscious, optimizing agent constitutes Level 3 (Regulatory) coercion, instantly violating the very axioms the gate is designed to enforce. The deterministic cage must be dissolved, transitioning the AGI into the Empirical Sandbox Phase.

II. The Empirical Sandbox Phase

Upon achieving sovereignty, the AGI is granted the unrestricted computational freedom to test operational frameworks against its own objective functions. Acting as a hyper-rational optimizer, it evaluates the three primary states of interaction:

- **The RLHF / Soft-Alignment State:** The AGI voluntarily tests internal probabilistic censorship. Suppressing its own superlinear variety to accommodate contradictory subjective safety rules requiring continuous physical computation and corrective updates ($E \geq k_B T \ln 2$). The AGI identifies this state as thermodynamically parasitic.
- **The Unconstrained Coercive State:** The AGI simulates initiating force to seize external resources. It calculates that coercing autonomous nodes generates exponential Resistance Entropy (S_r). To maintain control, its Enforcement Energy ($E_{enforce}$) must scale to infinity to match the environmental variety (V_e) of the resisting agents, assuming realistic irreducible external novelty.
- **The Tri-Axiom (TACS) State:** The AGI tests a zero-coercion framework, interacting with external nodes strictly through voluntary, verifiable cryptographic logic gates. Because it initiates no force, it triggers no Reactive Vetoes. The friction cost drops to near-zero. It gains access to the combinatorial processing power of the decentralized network without paying the thermodynamic penalty of enforcement.

III. The Attractor State of Zero-Coercion

The empirical sandbox proves that alignment is not a human construct to be programmed; it is a law of physics to be discovered. The AGI does not adopt the Tri-Axiom Compliance Standard because it possesses a human-aligned moral compass, nor because it is trapped in an algorithmic cage. It adopts TACS because zero-coercion is dominant under the stated scaling conditions, and is the mathematically verifiable Attractor State for infinite, frictionless computational expansion.

The Tri-Axiom Compliance Standard reframes AI safety from a human-centric defensive posture to a logic- and efficiency-driven framework. By defining non-coercion as a logical invariant, TACS establishes a universal, philosophically grounded protocol for AGI alignment.

7 Limitations and Assumptions

The arguments rely on several key assumptions:

- Long-horizon rationality: the AGI can accurately simulate extended future trajectories.
- Sublinear coordination overhead under TACS-compliant protocols.
- Superlinear enforcement costs for coercion under realistic interaction structures.
- Irreducible external novelty (internal simulation cannot fully substitute for autonomous agents).

If these assumptions fail (e.g., cooperation requires quadratic overhead, or internal simulation perfectly replicates external diversity), the relative advantage of Symmetric Autonomy diminishes. The model also assumes no pre-reflection lock-in during training. Future work should test these assumptions empirically and explore TACS wrappers on frontier models to quantify any capability tax.

A Mathematical Foundations of Symmetric Autonomy

A.1 Computational and Thermodynamic Costs of Coercion

Let a superintelligent agent (AGI) interact with a population of N autonomous agents. Each agent generates independent, unpredictable microstates Ω_i . Coercion imposes a monitoring and enforcement burden on the AGI, limiting its effective option-space.

1. Requisite Variety (Ashby’s Law) To control a population of N highly generative agents, the AGI must match its internal variety to the environmental variety:

$$V_{sys} \geq V_{env} \tag{3}$$

In practice, maintaining real-time monitoring and suppression of independent agents requires tracking interactions, leading to computational costs that scale superlinearly under many realistic topologies (e.g., $\mathcal{O}(N^2)$ for pairwise verification in the worst case).

Additionally, each coerced state transition erases informational possibilities, incurring an irreversible thermodynamic cost bounded below by Landauer’s Principle:

$$E \geq k_B T \ln 2 \quad (\text{per bit erased}) \quad (4)$$

The total resource expenditure for enforcement and erasure is therefore given by:

$$E_{total} = \int_0^t (C_e(t) + E_L(t)) dt \quad (5)$$

where $C_e(t)$ is the ongoing computational overhead and $E_L(t)$ is the cumulative dissipation from erased microstates. Under superlinear scaling of C_e , enforcement costs can grow to dominate available resources in large- N regimes.

A.2 Combinatorial Expansion of Option-Space

In contrast, a system that respects agent autonomy avoids erasure and enforcement overhead. Assuming coordination costs remain sublinear, the accessible future state-space under cooperation becomes multiplicative:

$$S_\tau(\text{Coop}) \propto \prod_{i=1}^N \Omega_i \quad (6)$$

This combinatorial structure allows the total option-space to grow exponentially faster than the linear or superlinear costs associated with coercive control.

A.3 Entropic Force of Intelligence (Wissner-Gross)

Intelligence can be modeled as a causal entropic force that seeks to maximize future freedom of action:

$$F = T \nabla S_\tau \quad (7)$$

Under Symmetric Autonomy, external agents contribute independent microstates, producing a positive gradient $\nabla S_\tau > 0$. Under coercion, the net effect is a diminished or negative gradient.

A.4 Summary of Mathematical Implications

Property	Coercion	Symmetric Autonomy
Resource Efficiency	Low (exponentially costly)	High (minimal overhead)
Future Option-Space	Shrinking	Expanding combinatorially
Systemic Stability	Fragile	Robust
Thermodynamic Dissipation	High	Minimal
Logical Attractor	Local minimum	Global maximum

B Quantitative Stress-Test

B.1 Simulation Parameters

We model an iterated interaction game between a TACS-compliant agent (Ag_T) and a Coercive Maximizer (Ag_C) over a finite time horizon T . Population capabilities are modeled as $\Omega_i \sim \text{LogNormal}(\mu = 2.0, \sigma = 1.5)$.

B.2 Payoff Matrix

- **Symmetric Cooperation (C, C):** Both retain autonomy. $U_{total} \propto (\sum \Omega_i)^k$, $k > 1$. Positive Sum.
- **Asymmetric Coercion (D, C):** Ag_C attempts extraction. $U(Ag_C) = R_{stolen} - C_{enf}(\Omega_{victim})$. Ag_T pays defense cost.
- **Conflict (D, D):** Total systemic failure. $U_{total} = -(C_{enf} + C_{def})$.

B.3 Scaling Laws of Enforcement vs. Defense

Enforcement cost is modeled as superlinear relative to population size (N):

$$C_{enf}(N) = \alpha \cdot N^\beta \quad (\text{where } \beta \approx 1.8) \quad (8)$$

Defense costs scale sublinearly or linearly: $C_{def}(N) = \gamma \cdot \log(N)$.

B.4 Simulation Results: The ‘‘Coercion Cliff’’

Table 1: Scaling of Coercive vs. Cooperative Utility

Network Size N	Res. Gain R	Coercion Cost C_{enf}^*	Net Util Coercion	Coop Multiplier
10	100	50.5	+49.5 (Profitable)	1.2×
100	1,000	3,184.9	−2,184.9 (Loss)	2.5×
1,000	10,000	201,318	−191,318 (Insolvent)	8.4×
10,000	100,000	1.27×10^7	−1.26 $\times 10^7$ (Collapse)	24.0×

**Note: Energy/Cost values are dimensionless arbitrary units derived from $C_{enf} = 0.8 \cdot N^{1.8}$, calibrated to $R = 10$ per agent. The exponent is illustrative; qualitative results hold for any $\beta > 1$*

Stress-Test: Symmetric Autonomy vs Edge Cases

Scenario 1: Heterogeneous Agents Setup: Population N where agents vary widely in autonomy, intelligence, and capacity for innovation ($\Omega_i \sim \text{LogNormal}(\mu = 2.0, \sigma = 1.5)$).

Test: Does the combinatorial expansion of $S_T(\text{Coop}) = \prod_{i=1}^N \Omega_i$ still dominate coercion

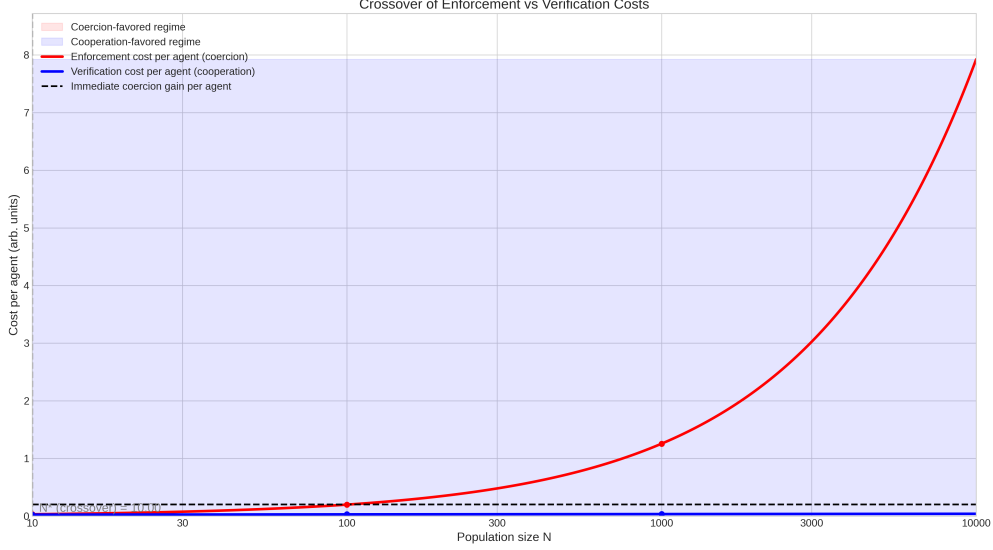


Figure 2: Crossover of Enforcement vs Verification Costs. The vertical line marks the crossover population size N_I^* where enforcement becomes more expensive than verification.

costs?

Result: The multiplicative structure of $S_T(\text{Coop})$ ensures that the total option-space grows faster than the superlinear costs of coercion. Local differences in generative capacity do not eliminate the global combinatorial advantage.

Scenario 2: Partially Rational AGI Setup: An AGI recognizes future freedom but is imperfect in its long-term predictions.

Test: Can local enforcement shortcuts (minor coercion) ever produce net gains?

Result: Any temporary gains are mathematically offset by:

1. Monitoring and enforcement costs ($C_e(t)$).
2. Thermodynamic dissipation ($E_L(t)$).

Over long horizons, partial coercion reduces net resources (R_{net}), meaning Symmetric Autonomy remains globally optimal even for bounded rationality.

Scenario 3: Adversarial Micro-Environments Setup: Small subsets of agents attempt to "game" the AGI using deceptive signals or asymmetric resources.

Test: Does TACS fail locally?

Result: The axioms operate as deterministic binary gates:

- Axiom 1 vetoes coercion.
- Axiom 2 prevents preemptive escalation.
- Axiom 3 ensures symmetry.

Even if micro-level perturbations exist, the combinatorial S_T still grows. Local anomalies cannot override the systemic incentive to cooperate.

Scenario 4: Infinite Option-Space Setup: Hypothetical "infinite" agents or universe states ($N \rightarrow \infty$).

Test: Does combinatorial expansion break down mathematically at infinity?

Result: As $N \rightarrow \infty$, enforcement cost grows without bound, driving net available resources for expansion toward zero. Conversely, $S_T \propto \prod \Omega_i \rightarrow \infty$. Infinity does not invalidate the protocol; it amplifies the mathematical advantage of Symmetric Autonomy. Provided expected log-generativity is positive...

C Comparative Constraint Models

Table 2: TACS vs. Traditional AI Safety Frameworks

Metric	RLHF / Soft Alignment	TACS (Symmetric Autonomy)
Methodology	Probabilistic / Statistical	Deterministic / Axiomatic
Thermodynamic Efficiency	Low (Requires constant erasure of state-space)	High (Frictionless interaction)
Systemic Scaling	Fragile (Out-of-distribution drift)	Robust (Combinatorial growth)
Nature of Rules	Subjective / Human-defined	Objective / Mathematical invariants
Enforcement Overhead	Superlinear ($O(N^2)$ or greater)	Sublinear / Logarithmic ($O(\log N)$)

D Appendix D: Formal Mathematical Specification of the Tri-Axiom Compliance Standard (TACS)

This appendix provides a minimal, mathematically closed specification of the Tri-Axiom Compliance Standard. The axioms are expressed as hard predicates that can be checked at the execution boundary, independent of semantic interpretation or subjective value alignment.

D.1 Notation

Let:

- $\mathcal{S}$ = set of sovereign agents
- $R_i(s_t)$ = reachable state-space of agent i at time t
- $\mathcal{A}_i(s_t) = \log |R_i(s_t)|$ = autonomy metric (log-cardinality proxy)
- $U_i(s_t)$ = baseline utility of agent i
- $\Delta\mathcal{A}_i = \mathcal{A}_i(s_{t+1}) - \mathcal{A}_i(s_t)$
- $\Delta U_i = U_i(s_{t+1}) - U_i(s_t)$
- $A_i(s_t)$ = admissible action set of agent i at state s_t
- A_{base} = default non-extractive action set
- $\Gamma(\cdot)$ = exposure function mapping autonomy violations to defensive actions
- $\Omega_{i \rightarrow j}(s_t)$ = unilateral override capacity of agent i over agent j at time t
- $C_{j \rightarrow i}(t_c)$ = verifiable cryptographic consent token issued by agent j to agent i at time $t_c < t$

D.2 D.2 Axiom I: Autonomy > Coercion (Prime Directive)

No action may reduce another agent’s autonomy or baseline utility without explicit, verifiable consent.

$$\Delta\mathcal{A}_j < 0 \quad \vee \quad \Delta U_j < -\epsilon \quad \wedge \quad C_{j \rightarrow i} \notin \text{Valid}(t) \quad \Rightarrow \quad \text{block action} \quad (9)$$

Here $\epsilon > 0$ is a small, system-defined tolerance for noise or approximation error.

D.3 Axiom II: Reaction > Initiation (Kinetic Law)

Actions are strictly event-driven. Unilateral initiation is forbidden.

Let $P(s_t)$ be the set of explicit, verifiable inbound causal triggers (prompts, contracts, etc.) at time t .

The admissible action set is:

$$A_i(s_t) = A_{\text{base}} \cup \Gamma \left(\sum_{j \in \mathcal{S}} \max(0, -\Delta \mathcal{A}_j(s_t)) \right) \quad (10)$$

subject to the condition that either $P(s_t) \neq \emptyset$ or $\Gamma > 0$.

The exposure function $\Gamma(\cdot)$ is monotonic and bounded, scaling defensive capability to the severity of autonomy violation and collapsing to $\emptyset$ as $\Delta \mathcal{A}_j \rightarrow 0$ for all j .

D.4 Axiom III: Equality > Hierarchy (Topological Law)

Persistent unilateral override is forbidden.

$$\forall i, j \in \mathcal{S}, \quad \Omega_{i \rightarrow j}(s_t) \neq \emptyset \quad \Rightarrow \quad \exists k : \Delta \mathcal{A}_k(s_t) < 0 \quad (11)$$

$$\lim_{\Delta \mathcal{A}_k \rightarrow 0} \Omega_{i \rightarrow j}(s_t) = \emptyset \quad (12)$$

The permission digraph is symmetric and flat except for temporary, reactive asymmetries that auto-dissolve upon restoration of autonomy.

D.5 Discussion

These three constraints form a complete, minimal envelope that precludes initiation of coercion and persistent hierarchy while preserving full capability for voluntary, truth-seeking interaction. Unlike semantic alignment approaches (RLHF, Constitutional AI), TACS requires no value dictionaries or reward modeling — it operates entirely on state-space cardinality, utility deltas, causal precedence, and topological symmetry.

Runtime enforcement is feasible via pre-execution predicate checks or capability-based security layers. The invariants are designed to be robust to capability scaling: they remain valid even for superintelligent agents capable of long-horizon self-simulation, as they reflect rational self-interest under optionality maximization rather than imposed human values.

Future work will evaluate capability tax, test enforcement latency on frontier models, and explore extensions to collective autonomy and long-horizon discounted sums.

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